Topic 1 Review · How to Solve It

Algebra I · Module 3 Topic 1: Linear Equations and Inequalities · Every review question, the fastest correct way, a check, and the trap to avoid.

The 7 moves for all of Topic 1

1. **Word problem → equation.** The amount that happens **per** week/item gets the x. The amount you **start** with or pay once is the plain number. y = rate·x + start.
2. **Distribute first.** Multiply the outside number by **everything** inside ( ), watch the signs, then combine like terms on each side.
3. **Variables on both sides.** Move the **smaller** x-term. The x stays positive, so you never have to flip an inequality.
4. **One, none, or infinite?.** Simplify both sides. Different x-terms → **one**. Same x-terms, different numbers → **none**. Identical sides → **infinite**.
5. **Inequalities.** Multiply or divide by a **negative** → **flip** the sign. Open circle for < >, closed for ≤ ≥. With x on the left, the sign points the way to shade.
6. **Formulas and forms.** Undo what is done to the letter you want, in reverse order. When you divide, divide **every** term.
7. **Always check.** Put your answer back in the original problem. Both sides must match (or the inequality must be true).

Question 1

Jayden has $40 in his bank account. He deposits $15 each week to save for new headphones. Which equation can be used to determine the amount that Jayden has saved, *y*, after *x* weeks?

> Answer: $y = 15x + 40$

🔍 **Spot it** "Each week" is a rate → it multiplies x. "Has $40" is the start → the plain number.

⚡ **Fastest way**

1. Rate: 15 per week → 15x
2. Start: 40
3. y = 15x + 40

✓ **Check** After 2 weeks: 40 + 15 + 15 = 70, and 15(2) + 40 = 70 ✓

⚠ **Trap** y = 15 + 40x swaps the rate and the start.

Question 2

What value of k makes the equation $5 - 4(k + 3) = 9 - 2(2k + 6)$ true?

> Answer: No solution

🔍 **Spot it** Parentheses on both sides and "No solution" is a choice → distribute and compare.

⚡ **Fastest way**

1. Left: $5 - 4k - 12 = -4k - 7$
2. Right: $9 - 4k - 12 = -4k - 3$
3. Same $-4k$, different numbers ($-7 \neq -3$) → **No solution**

✓ **Check** Once the k-terms match, stop. Just compare the numbers.

⚠ **Trap** Picking a number. No number can make -7 equal -3.

Question 3

What is the solution set for $3(4 - t) > -2t + 15$?

> Answer: $t < -3$

🔍 **Spot it** Inequality with a negative x-term → move the smaller x-term so you don't have to flip.

⚡ **Fastest way**

1. Distribute: $12 - 3t > -2t + 15$
2. Add $3t$ (the smaller x-term goes away): $12 > t + 15$
3. Subtract 15: $-3 > t$, which means $\mathbf{t < -3}$

✓ **Check** Try $t = -4$: $3(8) = 24$ and $8 + 15 = 23$, $24 > 23$ ✓

⚠ **Trap** $t > -3$: you divided by -1 and forgot to flip.

Question 4

Solve the equation shown.
$0.3(4x - 6) = -1.8x + 4.2$

> Answer: $x = 2$

🔍 **Spot it** Decimals → you can multiply everything by 10 to work with whole numbers.

⚡ **Fastest way**

1. Distribute: $1.2x - 1.8 = -1.8x + 4.2$
2. Add $1.8x$: $3x - 1.8 = 4.2$
3. Add 1.8: $3x = 6$ → $\mathbf{x = 2}$

✓ **Check** $0.3(8 - 6) = 0.6$ and $-3.6 + 4.2 = 0.6$ ✓

⚠ **Trap** Only multiplying 0.3 by 4x and not by the 6.

Question 5

Does the equation shown below have one solution, no solution, or an infinite number of solutions? Explain your reasoning.

$\frac{1}{3}(12x + 30) = 4(x + 2) + 2$

> Answer: Infinite solutions: both sides simplify to 4x + 10

🔍 **Spot it** "One, none, or infinite?" → simplify each side and compare.

⚡ **Fastest way**

1. Left: ⅓ · 12x + ⅓ · 30 = 4x + 10
2. Right: 4x + 8 + 2 = 4x + 10
3. Identical sides → **infinite solutions**

✓ **Check** Try x = 0: ⅓(30) = 10 and 4(2) + 2 = 10 ✓. Any x works.

⚠ **Trap** Getting 0 = 0 and writing "x = 0". 0 = 0 means *every* x works.

Question 6

Maya goes to a school store. She buys 4 notebooks that cost $1.75 each and some pens for $1.20 each. Write an equation for the total cost, *y*, when Maya buys *x* pens.

> Answer: $y = 1.20x + 7.00$

🔍 **Spot it** "Some pens" is the unknown → pens get the x. Exactly 4 notebooks is a fixed cost.

⚡ **Fastest way**

1. Notebooks: 4 × 1.75 = 7.00
2. Pens: 1.20x
3. **y = 1.20x + 7.00**

✓ **Check** 2 pens: 7.00 + 2.40 = 9.40, and 1.20(2) + 7 = 9.40 ✓

⚠ **Trap** y = 1.75x + 4.80: the x went on the notebooks instead of the pens.

Question 7

Rewrite the equation in slope-intercept form.

$2x + 5y = 15$

> Answer: $y = -\frac{2}{5}x + 3$

🔍 **Spot it** Standard form → slope-intercept: get y alone.

⚡ **Fastest way**

1. Subtract 2x: 5y = −2x + 15
2. Divide **every** term by 5
3. **y = −⅖x + 3**

✓ **Check** x = 5: 2(5) + 5y = 15 → y = 1, and −⅖(5) + 3 = 1 ✓

⚠ **Trap** Dividing only the x-term: y = −⅖x + 15.

Question 8

The formula for the volume of a rectangular prism is $V = \ell wh$. Solve the equation for h.

> Answer: $h = \frac{V}{\ell w}$

🔍 **Spot it** Formula with only multiplication → divide by everything that is not h.

⚡ **Fastest way**

1. h is multiplied by ℓw
2. Divide both sides by ℓw
3. **h = V/(ℓw)**

✓ **Check** ℓ = 2, w = 3, h = 4 gives V = 24, and 24 ÷ 6 = 4 ✓

⚠ **Trap** h = V − ℓw: you subtract to undo adding, not to undo multiplying.

Question 9

Solve the inequality, and graph the solution on the number line.

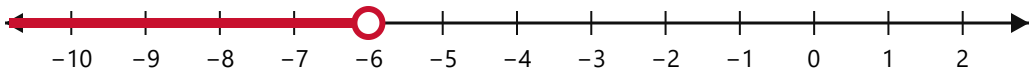
$2.5m - 3 < -1.5(m + 18)$

> Answer: $m < -6$ (open circle at −6, shade left)

🔍 **Spot it** Inequality with x on both sides → move the smaller x-term, then graph.

⚡ **Fastest way**

1. Distribute: 2.5m − 3 < −1.5m − 27
2. Add 1.5m: 4m − 3 < −27
3. Add 3: 4m < −24 → **m < −6**
4. Open circle at −6 (no "equal to"), shade left



✓ **Check** m = −7: −17.5 − 3 = −20.5 and −1.5(11) = −16.5, −20.5 < −16.5 ✓

⚠ **Trap** A closed circle. < does not include −6.

Question 10

Rewrite the equation $y - 4 = 3(x + 2)$ in slope-intercept form.

> Answer: $y = 3x + 10$

🔍 **Spot it** Point-slope → slope-intercept: distribute, then get y alone.

⚡ **Fastest way**

1. Distribute: y − 4 = 3x + 6
2. Add 4: **y = 3x + 10**

✓ **Check** x = −2: y − 4 = 0, so y = 4, and 3(−2) + 10 = 4 ✓

⚠ **Trap** y = 3x + 2: subtracting 4 instead of adding it.

Question 11

A triangle has a height of $(3n - 6)$ inches and a base of 10 inches. A rectangle has a length of 5 inches and a width of $(n + 4)$ inches. The area of the triangle is equal to the area of the rectangle. What is the value of n?

> Answer: $n = 5$

🔍 **Spot it** Two shapes with equal areas → write each area, set them equal.

⚡ **Fastest way**

1. Triangle: $\frac{1}{2} \cdot 10 = 5$ first, so $5(3n - 6) = 15n - 30$
2. Rectangle: $5(n + 4) = 5n + 20$
3. $15n - 30 = 5n + 20 \rightarrow 10n = 50 \rightarrow \mathbf{n = 5}$

✓ **Check** Height $3(5) - 6 = 9$: $\frac{1}{2} \cdot 10 \cdot 9 = 45$. Width $5 + 4 = 9$: $5 \cdot 9 = 45$ ✓

⚠ **Trap** Forgetting the ½ in the triangle's area.

Question 12 (extra practice)

Which equation has **no solution**?

> Answer: $3(x - 2) = 3x + 5$

🔍 **Spot it** Looking for no solution → you need the **same** x-term on both sides.

⚡ **Fastest way**

1. Distribute each choice
2. $3(x - 2) = 3x + 5 \rightarrow 3x - 6 = 3x + 5$: same $3x$, $-6 \neq 5$
3. $\mathbf{3(x - 2) = 3x + 5}$

✓ **Check** $3(x - 2) = 3x - 6$ is identical → infinite, not none.

⚠ **Trap** Choices with different x-terms always have exactly one solution.

Question 13 (extra practice)

What is the solution set for $-4x + 7 \geq 19$?

> Answer: $x \leq -3$

🔍 **Spot it** Negative in front of x → you will divide by a negative and flip.

⚡ **Fastest way**

1. Subtract 7: $-4x \geq 12$
2. Divide by -4 and flip: $\mathbf{x \leq -3}$

✓ **Check** $x = -4$: $16 + 7 = 23 \geq 19$ ✓

⚠ **Trap** $x \geq -3$: no flip.

Question 14 (extra practice)

The perimeter of a rectangle is $P = 2\ell + 2w$. Solve for w.

> Answer: $w = \frac{P - 2\ell}{2}$

🔍 **Spot it** Solve a formula for w → undo the adding, then the multiplying.

⚡ **Fastest way**

1. Subtract 2ℓ: $2w = P - 2\ell$
2. Divide every term by 2: $\mathbf{w = (P - 2\ell)/2 = P/2 - \ell}$

✓ **Check** $\ell = 3$, w = 2 → P = 10, and 10/2 − 3 = 2 ✓

⚠ **Trap** w = P/2 − 2ℓ: only the P was divided by 2.

Question 15 (extra practice)

A gym charges a $25 sign-up fee plus $8 per class. Lena can spend at most $90. Write and solve an inequality for the number of classes, c, she can take.

> Answer: $25 + 8c \le 90 \rightarrow c \le 8.125$, so at most 8 classes

🔍 **Spot it** "At most" → ≤. Fee paid once is the start; "per class" is the rate.

⚡ **Fastest way**

1. $25 + 8c \le 90$
2. Subtract 25: $8c \le 65$
3. $c \le 8.125$ → whole classes → **at most 8**

✓ **Check** 8 classes: 25 + 64 = 89 ≤ 90 ✓. 9 classes: 97 > 90 ✗

⚠ **Trap** Rounding up to 9. You can't go over $90.